

Firdovsi Rzaev

Software Engineer

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Profile

Software engineer and team lead with production experience designing institutional-scale web platforms — including learning management systems, research portals, and grant management infrastructure. Currently lead a software development team and contribute as in-house developer at Azerbaijan Technical University, where my systems are used daily by faculty, researchers, and students. Seeking an MSc/MASc and research-software-engineering opportunities at the intersection of **distributed systems**, **education technology**, and **research infrastructure**.

Education

Azerbaijan Technical University (AzTU)

Baku, Azerbaijan

Expected 2027

B.Sc. in Computer Science

GPA: **85.2 / 100** · Relevant coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, Computer Networks, Web Technologies, Distributed Computing.

Professional Experience

Software Developer Team Lead

2025 – Present

Dithari · Baku, Azerbaijan

- Lead a cross-functional engineering team delivering a full Learning Management System used daily by instructors and students for coursework, attendance, and assessment.
- Define system architecture, code standards, and review processes across frontend (Next.js, TypeScript) and backend (Spring Boot, FastAPI); mentor developers on testing, Git workflows, and production deployment.
- Architect role-based access control, multi-tenant data models, and REST/WebSocket APIs powering real-time classroom features (QR attendance, AI tutor, live syllabus).
- Own the DevOps pipeline: Docker containers, automated CI/CD with GitHub Actions, and Nginx-fronted deployments on Linux servers.

Instructor — Programming & Robotics

Nov 2025 – Present

Dithari Innovation and Sustainability Centre · Baku, Azerbaijan

- Teach programming and robotics, designing hands-on lessons that bridge fundamental computer science concepts with practical engineering skills.
- Develop curriculum and project-based exercises covering algorithmic thinking, software development workflows, and applied robotics; mentor students through end-to-end projects.

Software Developer

2024 – Present

Azerbaijan Technical University (AzTU) · Baku, Azerbaijan

- In-house developer for the university's digital infrastructure; designed and shipped **six production platforms** adopted at AzTU, including the Plan–Report system, E-Grant portal, Researchers portal, Majors portal, and the official university website.
- Integrated external academic data sources (Scopus, Web of Science, Google Scholar) to automate researcher profile and publication synchronization for the entire university.
- Worked directly with academic stakeholders — rectorate, deans, department heads, and research office — to translate institutional workflows into reliable software.

Selected Projects

Learning Management System (LMS)

Dithari Innovation and Sustainability Centre · Team Lead

Production platform for digital instruction — daily users.

- End-to-end LMS covering syllabus management, assignments, digital lesson workflows, and a student/teacher role system.
- Designed **QR-based attendance** with anti-spoofing checks, and integrated an **AI tutor** via LLM APIs to provide context-aware student support.
- *Stack:* Next.js, TypeScript, Tailwind, FastAPI / Spring Boot, PostgreSQL, Docker, Nginx, GitHub Actions.

E-Grant Portal — AzTU

Institutional research grant management system

Used by the university research office and external expert reviewers.

- Built the full grant competition workflow: candidate registration, project submission, document handling, expert assignment, blind evaluation, and final ranking pipeline.

- Designed role-based access for applicants, reviewers, and administrators; implemented audit trails for evaluation integrity.
- *Stack*: Spring Boot, PostgreSQL, Next.js, Docker.

Researchers Portal — AzTU

Centralized academic profile and publication system

Synchronizes publications across major academic databases.

- Integrated **Scopus**, **Web of Science**, and **Google Scholar** APIs into a unified publication ingestion pipeline with deduplication and identifier reconciliation (DOI / ORCID).
- Built researcher profile system with publication history, citation metrics, and institutional reporting.
- *Stack*: Spring Boot, PostgreSQL, REST integrations, Next.js.

Plan–Report Information System — AzTU

Workflow automation for academic departments

Replaces manual annual planning and reporting across the university.

- Automated yearly planning and reporting workflow with hierarchical approval chains, structured forms, versioned records, and exportable PDF reports.

Majors Portal — AzTU

Public-facing academic program explorer

Career and academic guidance for prospective and current students.

- Catalog of academic majors with structured **PLO/SLO** (Program / Student Learning Outcomes), syllabus access, and career guidance content.
- Designed information architecture and search to support student decision-making.

E-Commerce Platform — Buyology Trading FZ-LLC

Scalable commerce backend · Team Lead

Order management, product catalog, and admin operations.

- Led the engineering team building a scalable e-commerce platform with full order lifecycle, product management, and operations backend; defined the backend architecture, API contracts, and deployment topology.

AzTU Official Website

Software Developer

Primary public web presence of the university — contributed to development and maintenance, focusing on content systems and performance.

Upcoming Work

- **EduPlatform** — a nationwide education platform for Azerbaijan supporting online/offline courses, conferences, and an integrated instructor–student ecosystem, intended as shared educational infrastructure.
- **Giveaway Application** — a transparent, auditable raffle platform with provable randomness and anti-fraud safeguards.

Technical Skills

Languages	TypeScript, JavaScript, Java, Python, SQL
Frontend	React.js, Next.js, Tailwind CSS, REST/WebSocket clients
Backend	Spring Boot, FastAPI, Django, REST APIs, WebSocket services
Databases	PostgreSQL, MySQL — schema design, indexing, query optimization
Infra / DevOps	Docker, Linux, Nginx, GitHub Actions, CI/CD pipelines
Engineering	System design, backend architecture, role-based access systems, API integration, scalable web platforms

Research Interests

Research Software Engineering · Distributed Systems · Education Technology · AI-assisted Learning Systems · Data Systems · Research Infrastructure Systems

Leadership & Collaboration

- **Engineering team lead** at Dithari — responsible for architecture decisions, code review, mentoring, and delivery timelines.
- **Teaching & mentoring** — instructor for programming and robotics at Dithari Innovation and Sustainability Centre, designing curriculum and guiding students through hands-on technical projects.
- **Institutional systems ownership** at AzTU — shipped multiple platforms with day-to-day use by faculty, research office, and students; collaborated directly with deans, department heads, and the research office.

Publications & Research Work

Upcoming. Manuscripts in preparation on research-infrastructure software and AI-assisted educational platforms (working titles available on request).

Languages

Azerbaijani (Native) · English (Professional Working Proficiency) · Turkish (Conversational)